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Measurements of fuel mixture fraction oscillations of a turbulent jet non-premixed flame

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ABSTRACT

This work describes new type of combustion instability for which the 3-way coupling between mixing, flame heat release, and acoustics is modified by local buoyancy effects. Measurements of fuel mixture fraction are made for a non-premixed jet flame in a combustion chamber to assess the dynamics of mixing under imposed acoustic oscillations (22–55 Hz). Infrared laser absorption and phase resolved acetone-planar laser induced fluorescence are used to measure the fuel mixture fraction and then the degree of fuel/air mixing is calculated by determining the unmixedness. Results show acoustic excitation causes oscillations in the degree of fuel/air mixing at the driving frequency, which results in oscillatory flame behavior. This oscillatory flame behavior couples to the buoyancy and this in turn affects the mixing. Results also show that the mixing becomes less effective when the excitation frequency is increased or when the flame is present, compared to the non-reacting case. This work describes a key coupling mechanism that occurs when buoyancy is a significant factor in the flow field.

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1. Introduction

Combustion instabilities are pressure oscillations in a combustion chamber caused by coupling between thermo-acoustic and fluid-dynamic processes. Such oscillations can cause catastrophic damage in extreme cases. The interaction between vortices, sound waves and combustion heat release can lead to self-excited oscillations that can cause problems such as increased pollutant production, uneven burning, and, possible structural damage. Any unsteadiness in the process of combustion can be a source of pressure and velocity fluctuations [1,2]. Recently, environmental concerns have led to the use of lean premixed (LP) combustion schemes which reduce the amount of NO_x produced by the combustion system. The NO_x produced through the 'thermal or Zeldovich pathway' caused by the high temperature inside the combustors can be reduced by lowering the flame temperature. These lower temperatures can result in the occurrence of instabilities, since combustion in these systems tends to occur near the lean blow-out limits [3–5].

Several different studies have examined the physical phenomena involved in this process. A study of the acoustic coupling between fuel injectors and an applied acoustic field was carried out by Anderson et al. [6] that focused on cold flow experiments. Chen et al. [7] studied premixed flames designed specifically to simulate solid rocket propellants, and produced one-dimensional results at two forcing frequencies. Cohen et al. [8] studied the impact of mixing on the performance of a combustor by modulating part of the fuel flow for use in active control but did not make detailed observation of the dynamics inside. Davis and Chehroudi [9] also studied the governing parameters and consequences of acoustic waves imposed on the fuel jet. They performed experiments using a co-axial jet, and measured the temperature distribution and the jet core length for a jet undergoing an acoustic excitation (3 kHz) which was occurred close to the instability limits.

Theoretical and experimental studies of combustion instability have been conducted at Caltech for over 30 years. Culick [10,11] developed a theory that is based on the Rayleigh index. The coupled effects of acoustic forcing and combustion heat release on species concentration inside the flame zone were examined by Pun et al. [12,13] using OH-PLIF and chemiluminescence. This revealed that in the flame region the response of the combustion process under acoustic excitation is strongly phase-dependent, and this dependence was quantified. The flame base behavior and NO production rates were also studied [14,15]. Work by Kang et al. [16] focused on the measurement of swirl combustor dynamics, employing modern laser-based experimental techniques, and

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Nomenclature

σ	standard deviation	x	fuel–air ratio
σ_g	standard deviation of spatial deviation of fuel–air ratio (x)	x_{max}	maximum fuel–air ratio over space or time
σ_t	standard deviation of temporal deviation of fuel–air ratio	x_{min}	minimum fuel–air ratio over space or time
		$\langle x \rangle$	averaged fuel–air ratio over space or time

the subsequent calculation of the combustion dynamics. This effort extended the range of experimental forcing frequencies up to 400 Hz. The results [16] showed that the acoustics interaction with the flow field had a strong effect on the thermo-acoustic instability (the Rayleigh index) and created a periodic toroidal pattern in the Rayleigh index distribution that seems to be coincide with the toroidal vortex shedding present due to the Kelvin–Helmholtz instability in the flow field.

Acetone PLIF has been used to measure fuel concentrations and temperature [17,18]. Improvements have been made by Stufflebeam et al. [19], Thurber et al. [20,21], and Yip and Miller [22]. Lee and Santavicca [23] discussed a number of experimental diagnostic techniques for characterizing unstable combustion; the diagnostics include chemiluminescence emission, infrared absorption, and laser-induced fluorescence measurements (LIF).

The objective of the present study is to examine the effect of acoustic oscillations, which are sources of unstable dynamics inside the combustor, on the fuel/air mixing. This work also examines the link between flame behavior and mixing behavior. The current effort builds on previous work [13] by focusing on the fuel–air structure upstream of the flame front. To this end, two-dimensional images of the fuel–air distribution are obtained by imaging the acetone concentration. One-dimensional absorption measurements are seed to examine the behavior of fuel mixture fraction.

2. Experimental configuration

In this study, acetone was used as a fuel marker to visualize the fuel/air mixing region [24,25] directly upstream of the flame, and this is supplemented with time-resolved data acquired with an infrared absorption probe, as illustrated in Fig. 1.

2.1. Apparatus

A schematic view of the acoustic chamber is shown in Fig. 2. The acoustic chamber is made of a 30-cm-diameter aluminum central section which is 180 cm long and a stainless steel top that extends 50 cm to each side. The top of the chamber houses the loudspeakers and has a port to exhaust the product gases. The exhaust port is open to the atmosphere which creates an acoustically open exit condition for acoustic wave resonance. The bottom of the chamber is a closed-end acoustic boundary condition. The speakers are 30 cm in diameter and they are powered by 1000 W power amplifiers. A function generator (Wavetek 171) provides the input signal.

The burner is a jet-injection burner with the flame anchored approximately in the middle of the quartz tube. The fuel consists of 50% methane and 50% nitrogen with a trace amount of acetone seed. Air is entrained into the jet as the jet fluid moves through the burner. The fuel jet speed is approximately 20 m/s at the outlet of the 1.0-cm-diameter fuel port, yielding a jet Reynolds number of 10,000–20,000. The open-ended square cross-section quartz enclosure is 5.72 cm wide on each side and 11.43 cm tall.

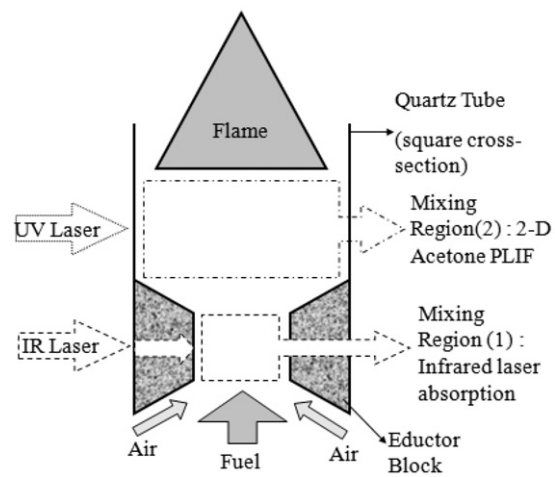

Fig. 1. Regions of interest; mixing is measured with IR laser absorption method at region (1) at the neck of the eductor block, with acetone PLIF at region (2) in the mixing zone.

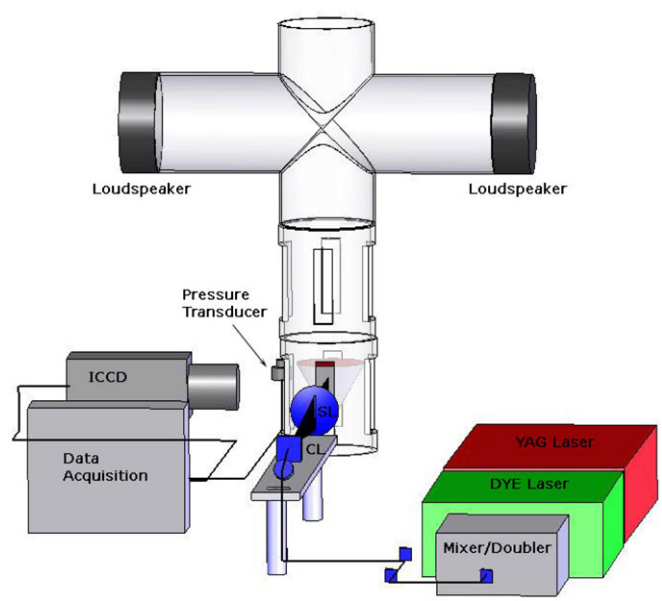

Fig. 2. Experimental configuration.

2.2. Infrared measurements

Temporal non-uniformities in mixture fraction were first measured by Fric [26] and Gulati and Warren [27]. These studies employed an argon ion laser to induce the fluorescence of NO_2, which had been seeded into the flow as a fuel marker. Mongia [28] and Hase and Kori [29] developed ways to make continuous measurement of methane concentration. The continuous nature of this technique allows sampling rates of several kilohertz with no additional chemical markers. The present study used a replica of

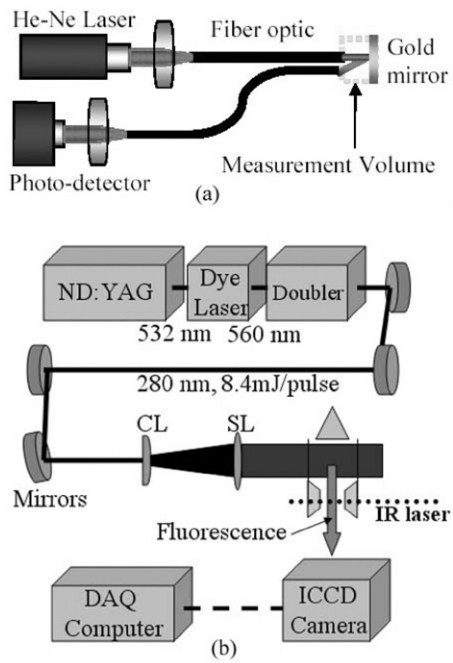

Fig. 3. Schematics of (a) infrared laser absorption method and (b) acetone PLIF.

Mongia's probe in order to make single-point measurements; the probe as shown in Fig. 3a.

The infrared laser probe assembly consists of a He–Ne laser which produces an infrared laser beam with wavelength of 3.39 μm (Fig. 3a). The laser beam is channeled down an optical fiber and into the measurement volume, through the gas mixture, reflects off of a gold, front-surface mirror, passes once more through the measurement volume, enters a receiving optical fiber, and is routed to a photo-detector. The measured extinction ratio yields (after calibration) the methane concentration in the control volume.

2.3. Acetone PLIF

Acetone PLIF is a widely used technique for measuring distributions of species concentrations [30–32] and for flow visualization purposes [33]. An intensified CCD camera was used for the PLIF imaging, while a data acquisition board (NI PCI 6014) along with a pressure transducer (PCB 106B50) was used to record the pressure, laser intensity, and ICCD gating signals. The image is captured on an intensified CCD camera with 300 by 225 pixels mapping to a 5.5 by 4.2 cm image plane area. Images are taken at random pressure phases, with the camera gating signal being recorded by the data acquisition system along with the pressure signal so as to enable appropriate post-processing.

The laser beam at 532 nm is used to pump a dye laser which operates at 560 nm. The output of the dye laser is then frequency doubled to 280 nm for excitation of acetone with the fluorescence signal expected in the range of 350–550 nm.

3. Results

As a measure of the degree of mixing, the unmixedness factor is employed in the present work. The unmixedness [34] depends on the variance of mixture fraction fluctuations σ_g^2. The Housdorf relation relates the maximum possible value of σ_g^2 to the average fuel–air ratio $\langle x \rangle$:

$$\sigma_{\max}^2 = \langle x \rangle \cdot \left(1 - \langle x \rangle\right), \tag{1}$$

so values of σ_g^2 are normalized by this maximum value.

3.1. Temporal fluctuation of fuel concentration

The temporal unmixedness employed here is defined as:

$$U = \frac{\sigma^2}{(x_{\max} - \langle x \rangle) \cdot (\langle x \rangle - x_{\min})}, \tag{2}$$

where σ is the standard deviation of the fuel concentration at one point over a number of successive measurements; x is the fuel concentration (mole-fraction); and $\langle\,\rangle$ is the average at a specific pixel (spatial location). When the temporal unmixedness (U) is zero, the fuel is completely mixed and is homogeneously distributed.

It can be seen in Fig. 4a that large values of temporal unmixedness occur in the mixing layer, at normalized radial locations that are 30–60% of the total radial distance. This is where shear-induced mixing between the jet and the entrained air is expected to occur. There is little variation in the overall structure of unmixedness distribution with respect to the varying acoustic forcing frequencies, and all plots indicate the same general mixing layer structure. In Figs. 4b and 4c, lighter shading indicates a region with a larger values of unmixedness, thus marking the region of high fuel concentration fluctuation. Further downstream from the jet nozzle (Fig. 4b), the shear layer becomes wider. The low values of temporal unmixedness in the core region (located between 0 and 30% of the distance from the center), is due to the large and relatively uniform fuel concentration. Cross sectional views cut from Fig. 4 at longitudinal distances of 1–3 cm are shown in Fig. 5.

Fig. 5a shows the temporal unmixedness for non-reacting case at three vertical heights above the eductor block while Fig. 5b shows the same for reacting cases. There is little temporal variation for mixing in the non-reacting case in both the radial and vertical location. On the other hand, Fig. 5b shows a relatively strong peak at the 50% radial location, and the peak slowly drops as the flow travels vertically upward.

This seems to be a direct result of buoyancy. Simple calculation based on a 2-D FLUENT model shows that, for reacting cases, the flow velocity in the downstream (upward) direction (opposite to that of gravity), mass-averaged within the mixing zone, is larger than the non-reacting flow velocity in the same zone. This zone spans from the centerline (0%) to about 60% of the radial distance to the outer edge (Fig. 6). The velocity distribution in Fig. 6a shows that the high velocity core of the reacting flow is widened compared to that of the cold flow, and this can be shown more clearly in Fig. 6b. It is evident that for radial locations of 30% or farther out, the mass-weighted velocity profile is larger for the reacting flow. The temporal unmixedness distribution pattern for the reacting flow shows that the shear mixing zone is formed approximately in the vicinity of 30 to 60% of the radial distance which agrees with Fig. 4. The profile also agrees with the unmixedness pattern seen in Figs. 4 and 5. So even in a system which should be primarily a jet-driven flow, the effect of buoyancy due to combustion still appears to have a significant impact.

This means that at the same vertical location, the distribution of unmixedness for the reacting flow cases will resemble the distributions at the lower vertical locations (which correspond to earlier times) for the non-reacting cases. This is evidenced by the fact that the non-reacting cases are mostly mixed by the time the flow reaches the imaging area while the reacting flows are still undergoing mixing. The characteristic time of the mixing should be greater than that of the convection for the current conditions. It also appears that the stabilizing presence of the flame due to gas expansion also slows down the mixing process.

Qualitatively, while the fuel/air mixing maintains the same structure downstream, but the measured gradients decrease. Fig. 7

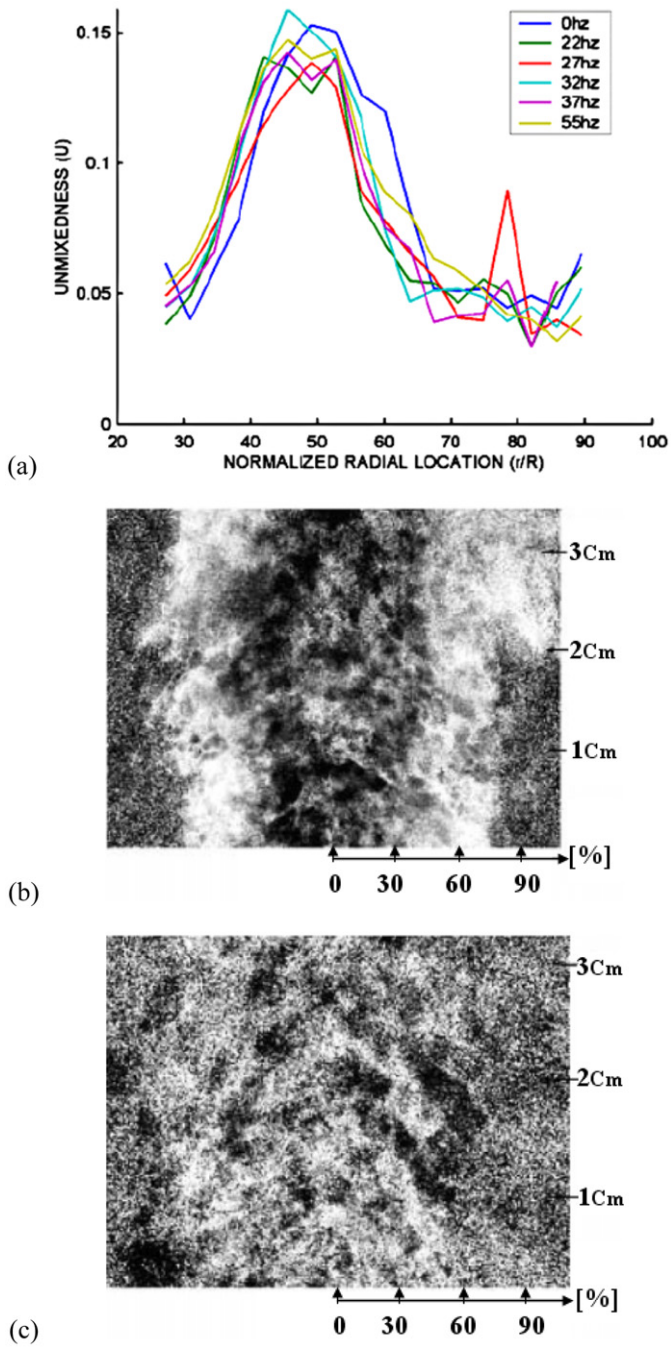

Fig. 4. (a) Temporal unmixedness factor at the neck of the eductor block defined by Eq. (1) and 2-D maps (adjusted by normalization) from PLIF measurements at 37 Hz, (b) reacting case and (c) non-reacting case (for b and c, higher values of temporal unmixedness factors are represented by brighter shading).

shows the fuel concentration distribution downstream, under the same condition as those described in Fig. 5, and normalized by the maximum value of the reacting case. As the flow convects downstream the degree of fluctuations in fuel/air mixture fraction decreases such that smaller values of unmixedness are observed in the outer flow region (60–80% of the radial distance from center). It is evident that acoustic forcing has a significantly weaker effect on the mean methane concentration than does the presence of combustion. Both reacting and non-reacting cases display the degree of self-similarity that would be expected of jet flows.

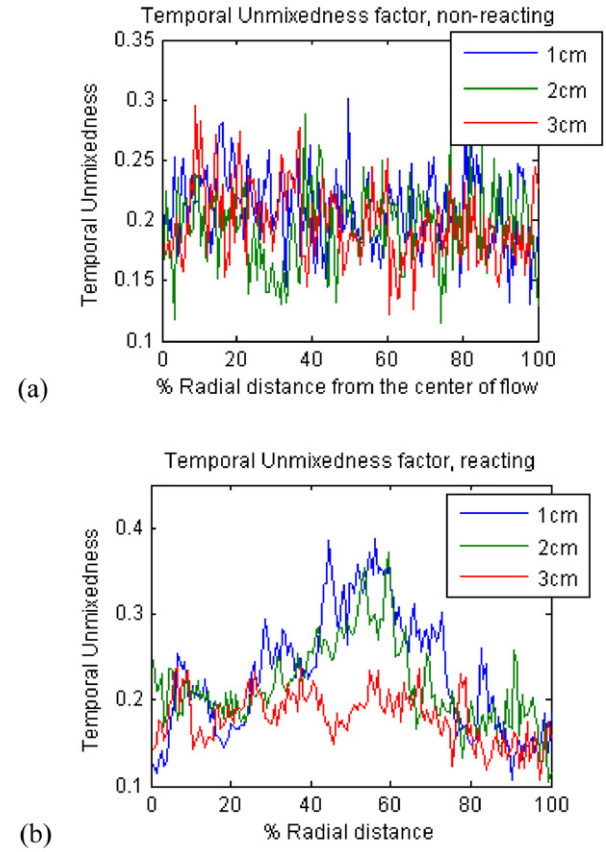

Fig. 5. Temporal unmixedness factor at each location with excitation frequency 37 Hz for (a) non-reacting and (b) reacting flow at 1, 2, 3 cm above the eductor block.

3.2. Phase-resolved behavior of mixing

The global unmixedness here is defined as

$$U_g = \frac{\sigma_g^2}{(1 - \langle x \rangle_g) \cdot \langle x \rangle_g}, \tag{3}$$

where σ_g is the standard deviation and $\langle x \rangle_g$ average of fuel concentration over the entire 2-D region. Previously the temporal unmixedness factor was described which requires that data are collected at many times at one point. The global unmixedness, in contrast, has a single value for each 2-D image, and this value is given by Eq. (3).

Phase dependence of the mixing behavior is shown in Fig. 8. Colors tagged with positive values indicate regions with higher than average fuel concentration (red and yellow), while the negative-valued contours indicate regions with lower than the average concentration (dark and light blue). The domain size shown here is 55 mm by 42 mm and the images underwent the background substitution. The values indicate the relative deviations of fuel concentration from the mean fuel concentration at the locations. The indicated phase is that of excitation acoustic wave. For 37 Hz, the highest overall fuel concentration occurs at the pressure node (0, 180 degrees), then decreases subsequently at 30(210) and 60(240) degrees an continues to decrease approaching the pressure anti-nodes at (90, 270 degrees).

Fig. 9 is a 3-D view of the power density spectrum; the power is plotted versus the frequency at each radial location. This shows that there is a very strong peak in the mixture fraction oscillations in response to an imposed acoustic field. This particular plot corresponds to reacting flow and a frequency of 32 Hz.

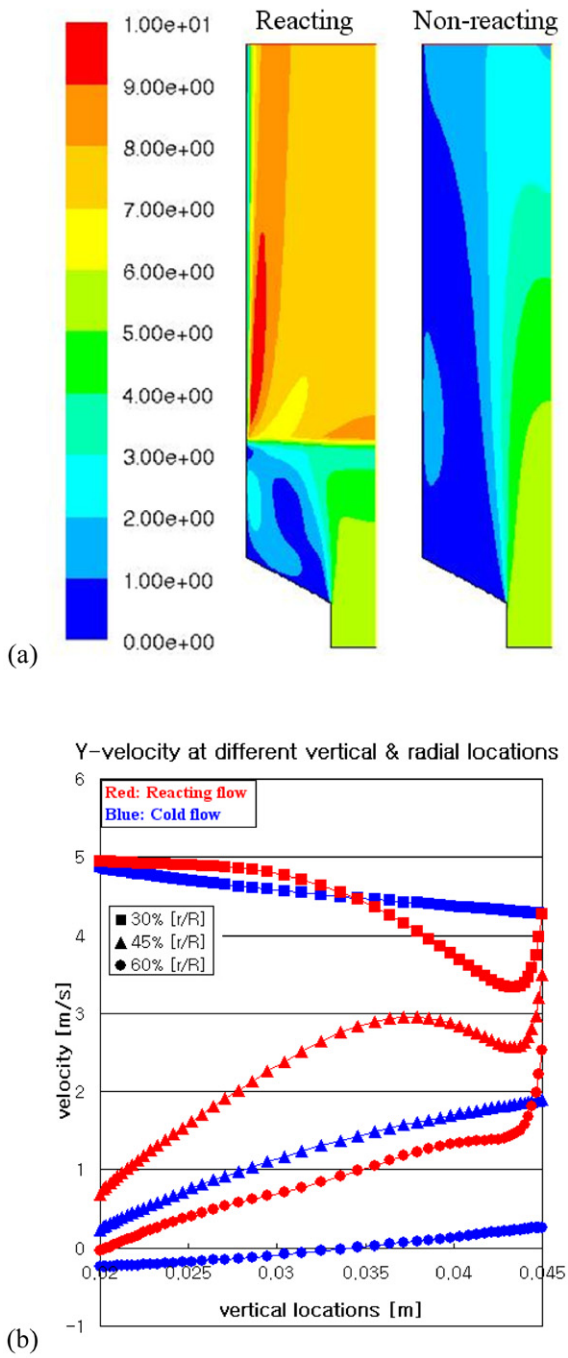

Fig. 6. (a) Mass weighted vertical velocity distribution in the mixing zone, and (b) the vertical velocity distribution at three different radial locations ($r/R = 30$, 45, and 60%) along the vertical direction.

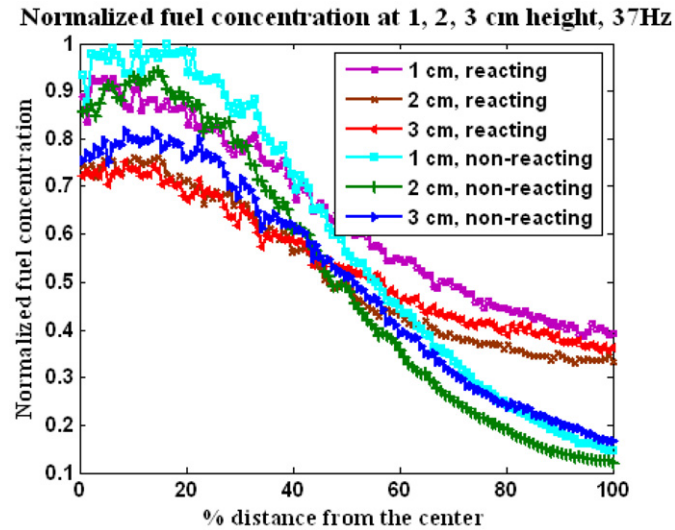

Fig. 7. The fuel concentration distribution.

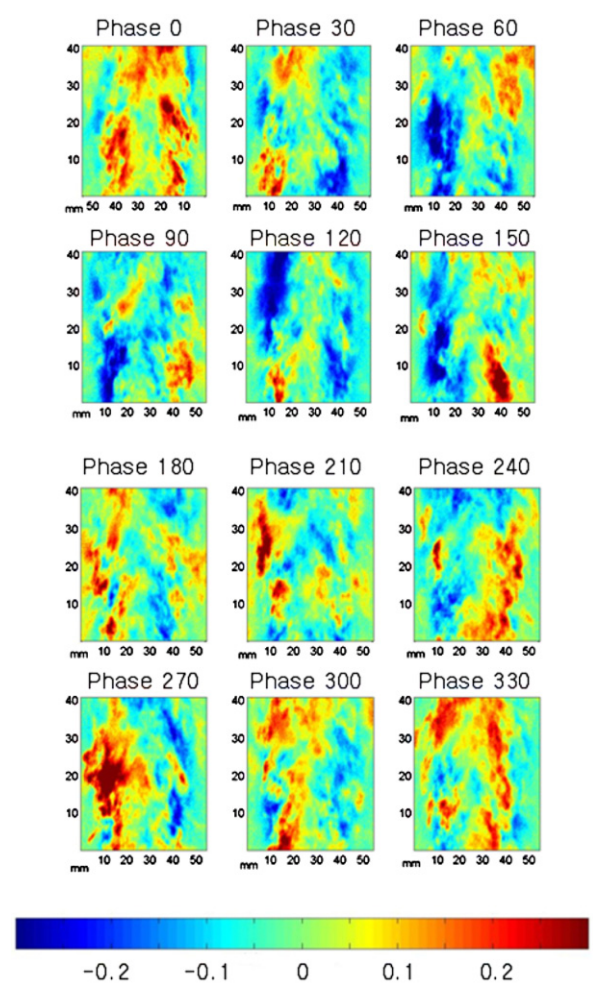

Fig. 8. Distribution of fuel concentration by phase at 37 Hz, reacting flow case.

Of particular interest is the very strong peak at 32 Hz, the driving frequency for this case. Similar peaks are evident in all of the acoustically forced cases, corresponding to the driving frequency. This indicates that strong coupling occurs between the acoustic field and the mixing layer.

Also, since no strong frequency preference is observed in the outer portion of the flow, the coupling is not a strong function of the vortex shedding from the fuel nozzle exit. The vortices are expected to be larger in size and play a role in producing oscillations in the local equivalence ratio in the outer flow region.

In Fig. 10a, the phase-resolved unmixedness at each phase for 37 Hz is shown. The curve was fitted by eighth order polynomial with 1-σ error bounds (68%) indicated in dots along with raw data points. As shown, the overall value of unmixedness is far greater for the reacting case, while the magnitudes of unmixedness fluctuations are about the same. The first mode of unmixedness fluctuation is shown by the curve for the 'non-reacting' case in Fig. 10a. The second mode rides the first mode as shown by the

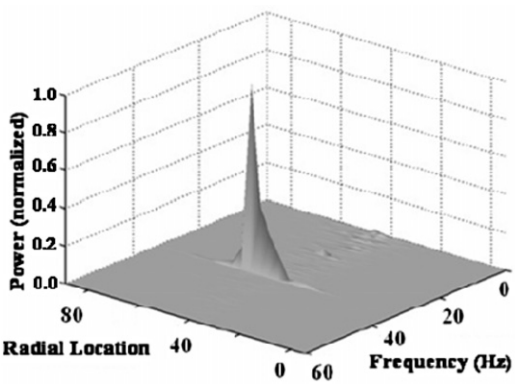

Fig. 9. Power density spectrum of fuel mixture fraction for 32 Hz, reacting flow.

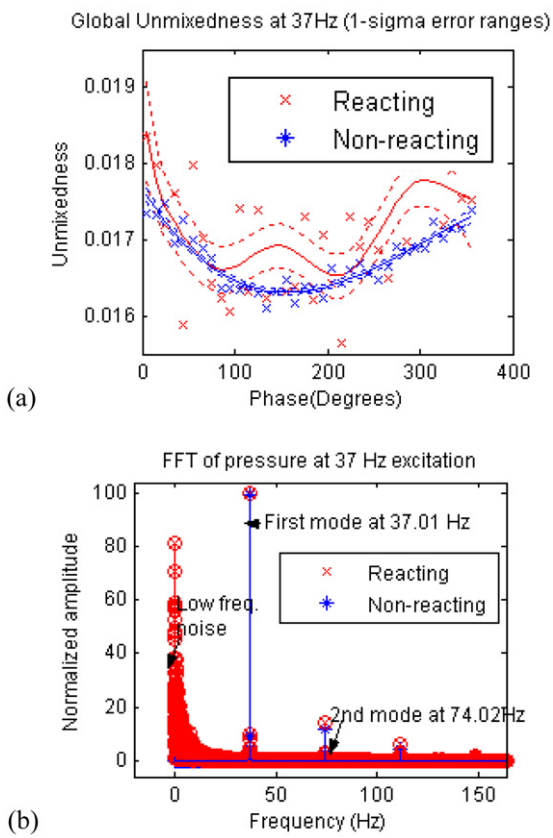

Fig. 10. (a) Behavior of mixing in term of global unmixedness (Eq. (2)) at 37 Hz of excitation with offset of 0.012 added to non-reacting case. Dotted lines indicate 1-σ error bounds (68%). (b) Measured pressure signal at 37 Hz of excitation.

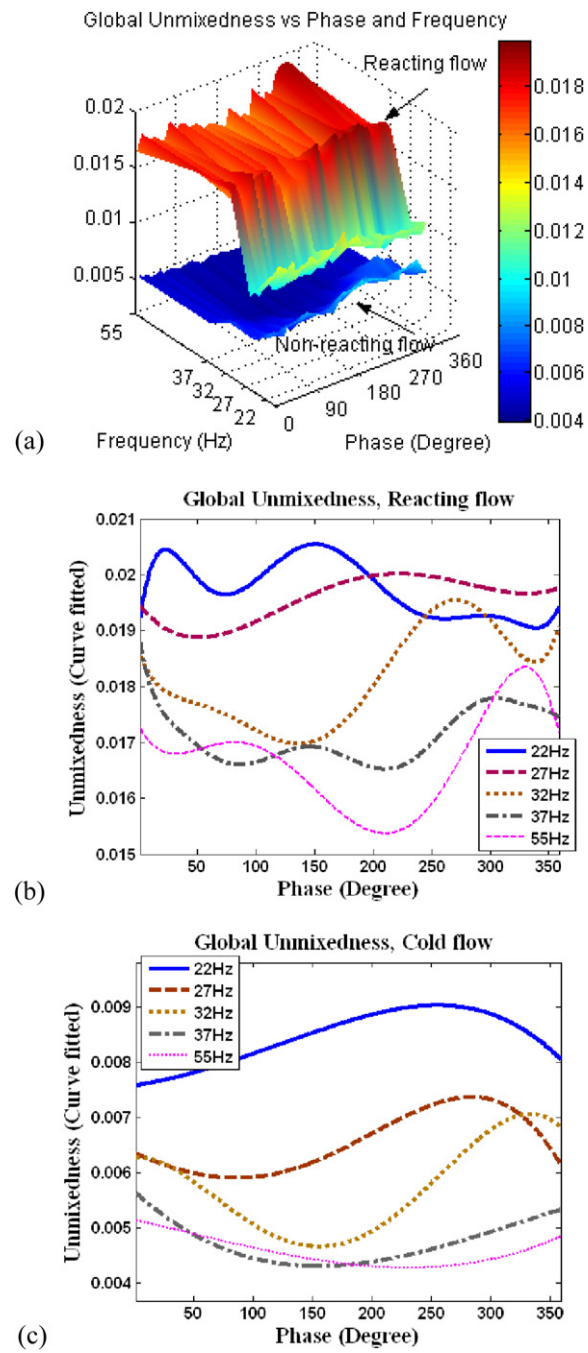

Fig. 11. (a) 3-D representation of global unmixedness vs excitation frequency and phase, and (b) reacting cases with offset added to 22 and 27 Hz, (c) non-reacting cases. (b) and (c) are curve-fitted by polynomials of order 8.

curve for the reacting case. This is observed at all other excitation frequencies as well. It seems that the frequency of the second mode is about two periods (or more) per 360 degrees thus twice the excitation frequency.

Combustion causes a region of irregular pressure fluctuations that is not seen in the non-reacting cases (see Fig. 10b) due to the generation of sound waves by the irregular burning of the turbulent flame. This phenomenon is sometimes called the '$1/f$ error,' and is observed in a wide variety of experimental work in the field of random processes. The amplitude of waves generated by the turbulent burning of fuel is significant and is due to fluctuations in the local equivalence ratio in the flame area, which itself is caused by irregularities in the fuel/air mixing upstream of the flame.

The global unmixedness is plotted versus excitation frequency and phase during one cycle in Fig. 11a. Significantly less mixing, based on the values of global unmixedness, is observed in the presence of the flame. The mixing at higher frequencies (32–55 Hz) is much more affected by the presence of flame than at lower frequencies (22, 27 Hz). This can be understood because the flame creates hot products that are accelerated upwards by buoyancy. This upward acceleration will naturally draw the jet downstream, so at the same downstream location, it is reasonable to expect to see a mixing profile that is analogous to a profile that corresponds to an earlier time for the non-reacting cases.

Fig. 11a indicates that the combustion process couples to acoustic excitation and produces additional changes to the mixing be-

havior. The reacting cases show an increase in unmixedness as the excitation frequency increases, while the non-reacting cases show the opposite tendency. The mixing behavior also is affected by the phase of excitation, though not so much as by the presence of combustion itself. Figs. 11b and 11c show how mixing behaves according to the excitation frequency and phase. All curves are fitted by polynomials of 8th order. For the reacting flow cases (b), the curves for the results at 22 and 27 Hz are shifted up for easier comparison of their qualitative behavior. Direct comparison of the curve shapes clearly shows that in reacting flow cases, the mixing fluctuates at higher frequencies than the imposed acoustic oscillations while still oscillating mainly at the imposed frequencies. On the other hand, non-reacting flow cases show only the oscillations at the imposed acoustic frequencies. These differences in the mixing behavior must arise due to the difference in thermo-acoustic coupling. The flame behavior is, in turn, affected by fluctuations in the mixing and this then couples back to the acoustic field.

4. Conclusion

This work has demonstrated the existence and has characterized the behavior of a new combustion instability coupling mode that relates weak acoustic forcing to several factors: mixture fraction oscillations, flame oscillations and buoyancy forces. The mode is characterized by a 3-way coupling between the acoustics, mixing, and flame heat-release with the buoyancy acting to modify the frequency and phase relationship of the coupling. The buoyancy acts to accelerate the flow field and hence reduce the amount of time available for mixing. This results in a less well mixed flow and stronger mixture fraction oscillations. Cold flow tests cannot identify this mode without changes to the testing and analysis methodology.

The phase-resolved global unmixedness measurements indicate the presence of the primary mode of oscillation, which occurs at the same frequency as the excitation. Conversely, the unmixedness oscillations are phase shifted from what is seen for the flame base behavior, and the amount of shift varies for different frequencies. It is clear that the acoustic forcing causes a strong periodicity in the mixing layer at the driven frequency for all cases. However, higher harmonic frequencies are present only in the reacting flow cases. It appears that these oscillations in the unmixedness are then the main cause of fluctuations in local equivalence ratio in the flame region, which, in turn, induces the periodic flame motion observed in the previous work [12,13]. The periodic flame motion creates large values of local Rayleigh index which drives the acoustic field.

The temporal unmixedness pattern reveals that the breakdown of the fuel/air distribution structure is slower in the reacting flow cases because the flow field is modified by the buoyant acceleration due to the high temperature of the hot product gases, resulting in slower mixing due to the loss of mixing time and wider variations in the mixing zone. In general, the mean flow patterns keep the same structure downstream of the mixing region and as the flow progresses downstream, subsequent mixing reduces the variation in the distribution of fuel/air mixing in both reacting and non-reacting cases.

Acknowledgments

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